# MINI PROJECT REPORT

Submitted to the faculty of Engineering and Technology VI Semester B.Tech

(Autonomous Batch)

*A Mini Project report on*

**IMAGE COLORIZATION USING CNN**

BY

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**KAKATIYA INSTITUTE OF TECHNOLOGY & SCIENCE**

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# CERTIFICATE

This is to certify that **SHARVIKA REDDY** bearing roll no: **B23DS047** of the VI Semester B.Tech. Computer Science and Engineering (AI&ML) (Autonomous) has satisfactorily completed the Mini Project dissertation work entitled **"IMAGE COLORIZATION USING CNN"**, in partial fulfillment requirements of B.Tech degree during the academic year 2025-26.

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# ABSTRACT

A web-based Inventory and Billing Management System is a helpful tool for retail stores and small businesses to run their day-to-day operations more efficiently by enabling them to track their inventory levels and generate invoices. Managing inventory and generating invoices manually creates opportunities for error and data loss, making business operations less efficient. The Inventory and Billing Management System gives users a way to store product information digitally, monitor their inventory levels, and create invoices quickly and accurately.

The Inventory and Billing Management System was created using various modern web technologies such as HTML, CSS, JavaScript and React.js for the frontend; Node.js and Express.js for the backend; and MongoDB for the database. The Inventory and Billing Management System has many key features such as product management, inventory tracking, automatically generating invoices, real-time transaction updates to stock inventory levels, and many more.

The primary benefit of this system is to reduce manual efforts; increase record accuracy; and improve the overall efficiency of a business. By automating repetitive tasks within the retail environment, the system allows the retail owner to spend their time on other business needs while minimizing the chance of making errors and allowing their data to be stored in an orderly manner. This project demonstrates how digital solutions like the Inventory and Billing Management System can help small businesses become more efficient and productive.

The level of protection of data within this system has more protection than the level of protection of data available to organizations that have direct access to that data. You do not need an IT background to use this system, because it is an easy to use system. You can add many features (for example, reporting / analysis) to the system that will assist the user(s) in making good choices around sales. Furthermore, because the system has a scalable architecture, it provides the organization with the ability to grow its business without having to build a new system or significantly change the existing system to support growth. The success of this project is an outstanding and compelling demonstration of the many ways in which organizations can utilize digital tools to grow their business and to do so in a sustainable manner.

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# CHAPTER - 1

# INTRODUCTION

## 1.1 INTRODUCTION

Currently, in this digital age, there is a need for companies to have effective management processes for daily operations including inventory management, billing, customer management, sales management, among other. Smaller scale companies including retail stores, supermarkets, pharmaceutical stores, warehouses, electronic stores, among others experience high volumes of products and customer interaction on a daily basis. Handling these processes manually using paper files, books, spreadsheet, or even using calculators becomes challenging due to growth in business operations.

Inventory management includes having detailed stock management processes including managing the availability and the updating of the quantity of each product. On the other hand, billing management is the process of issuing bills, totaling amounts, recording transactions, and purchasing goods by customers. When the two management processes are done manually, many difficulties will arise including inaccurate stock quantities, late billing, wrong calculations, among others.

Therefore, in order to solve the above difficulties, a web based software application known as Inventory and Billing Management System was created. The management software includes automating stock management, billing, invoicing, customer management, and creating reports.

This project was developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. It was chosen since it makes it possible for one to use JavaScript on the front end and back end sides, thus making the whole development process easy, fast, and scalable.

For the development of the frontend of the system, the React.js framework was used to ensure that we have a responsive, dynamic, and user-friendly interface. For the backend side, the Node.js and Express.js frameworks were used to deal with APIs, business logic, and authentication. We have the MongoDB database to manage the information concerning products, invoices, customers, and users.

Some of the modules included in the project include secure logins and authentications, product management, inventory, customer management, billing and invoicing, reporting dashboard, and role-based permissions. For the admin users, they will be able to manage all aspects of the system, while staff users will only be able to access specific features. With the help of the billing component, customers will be able to search products, place their selected products into the cart, make automatic calculations, include taxes and discounts, and generate invoices immediately. Every time a purchase is made, the inventory count updates itself instantaneously.

For this project, the frontend was hosted on Vercel, the backend was hosted on Render, and the MongoDB database was hosted online on MongoDB Atlas.

Inventory and Billing Management System offers fast billing, minimizes the risk of errors, boosts productivity, improves customer service, and aids effective decision-making with help of reports and analytics.

## 1.2 PROBLEM STATEMENT

Some of the smaller and middle-level firms are relying on the traditional manual system of doing business. In this, they use pen and paper bills, paper-based records, diaries, calculators, and spreadsheets for their accounting and billing process. Though this might be adequate for small-scale businesses, as the business expands, the use of manual systems becomes less feasible.

The following are some of the issues that manual systems bring about: inaccurate inventory, delay in billing, duplicate items, wrong calculations, omission of customers' particulars, and failure to maintain transaction records. During peak business times, manual billing tends to slow down service delivery by increasing waiting time for customers. The other problem is that when there is no real-time stock tracking, stock shortages might occur.

Other important problems associated with manual systems are security and data handling. In this regard, business records stored in diaries and paper files are susceptible to damage, theft, or loss. Likewise, spreadsheet systems need manual inputting of data.

There are many pre-existing software options for billing or enterprise resource planning systems that can be utilized; however, these are costly, challenging to customize, and highly complex for small and medium-sized companies.

Thus, there is a necessity for a reliable, user-friendly, precise, and cost-effective digital solution that can handle inventory management, billing, invoicing, customer handling, and reporting all in one place.

To address these challenges, an Inventory and Billing Management System has been designed using the MERN stack. The proposed solution offers up-to-date inventory tracking, speedy billing, safe login authentication, user-specific authorization, and informative reports.

## 1.3 OBJECTIVES

The primary goals of creating the Inventory and Billing Management System are:

- Automation of inventory management tasks through precise monitoring of the quantity of stock, management of product information such as name, price, quantity, and category, and updating of stock in real-time.
- Streamlining of the billing process through fast creation of invoices and conducting sales transactions without much paperwork.
- Digital record keeping of the stock of the products and automated reduction of stock when there is a sale.
- Minimization of unnecessary work through automation of manual tasks in billing and inventory management.
- Secure storage and management of customer information in the system such as the customer's information and their past purchases.
- Efficient searching and filtering for customers and products to enhance retrieval of information from the database.
- Creation of reports on sales, billing, and inventory for analysis by management.
- For implementing the use of JSON Web Tokens (JWT) for secure user authentication and bcrypt to encrypt passwords.

- For the implementation of role-based access control where we can create roles like Admin and staff with unique access rights based on their respective roles.
- For implementing a user-friendly and responsive interface with the help of React, allowing for easier usage by Admin and Staff users.
- For building a scalable application by employing the MERN Stack framework, which consists of MongoDB, Express.js, React, and Node.js.
- For speeding up the billing process, increasing customer satisfaction, and improving employee productivity.
- For implementing a scalable solution on the cloud where our frontend will be deployed on Vercel, backend on Render, and the database on MongoDB Atlas.
- For achieving better efficiency and cost-saving measures.

## 1.4 OUTCOMES

The successful application of the Inventory and Billing Management System led to several benefits that will significantly improve the efficiency and productivity of the business operation processes. The system sales report generation.

- Fast and efficient billing operations will be guaranteed as the system will generate invoices fast and accurately.
- Accurate inventory management will be achieved since the product inventory will be digitalized, and quantities will be automatically updated after each sale.
- The project eliminates manual activities, paperwork, and frequent human mistakes associated with manual accounting.
- Data organization will be enhanced through the storage of product records, customer data, invoices, and other important information in the central database.
- Data safety will be ensured as the system will provide secure access based on JWT and bcrypt password encryption.
- The system uses role-based access control, meaning only authorized users can access different parts of the software.
- Customer management will be facilitated due to customer record storage.

- Useful sales and inventory reports will be generated, which will allow the owners to perform analysis.
- A responsive user interface developed with React framework ensures great UX and convenient usage.
- Full-stack development using the MERN stack technologies has been successfully completed.
- Deployment of the project has been successful via Vercel for hosting the frontend, Render for hosting the backend, and MongoDB Atlas for hosting the database, thus enabling access to it through the Internet.
- In general, this system contributes to productivity, minimizes operational expenses, improves client satisfaction, and enables future growth.

## 1.5 LITERATURE REVIEW

Literature reviews provide a critical overview of previous studies and existing solutions pertaining to the main subject of a proposed project. This will help identify what new technology will be utilized, what weaknesses existed in previous solutions, and the gap for additional solutions exists. For the Inventory and Billing Management System, existing literature was reviewed regarding managing inventory, billing, invoicing, and cloud computing.

**Study 1: Review of Historical Literature on Inventory Management Systems**

According to Madamidola et al. (2024), there are numerous types of inventory management systems, which are increasingly important within current companies as they provide companies with the capability of performing efficient inventory and billing management. It was identified in the study that many of these companies continue to rely on older systems, resulting in errors, delays, and inefficient inventory management processes. The authors of the study concluded that there is a need for updated integrated inventory and billing systems.

**Study 2: Implementation of Information Systems in the Real World**

The aim of the research conducted by Supapsophon et al. (2003) was to develop an electronic inventory and billing management system for manufacturing businesses for the purpose of replacing manual inventory and billing record-keeping.

The study concluded that electronic information systems resulted in more efficient inventory management, eliminating the need for large amounts of paperwork, and increasing the efficiency of businesses.

### Study 3: Development of Web-Based Online Billing and Invoice Management System for Small Businesses

Talit Bakhit conducted a study dedicated to developing an online billing and invoice management system for small firms. The author aimed at enhancing the process of invoicing, billing, and transaction management via a website. The results indicated that online billing is more convenient and effective than traditional paper-based billing systems.

### Study 4: Cloud-Based Inventory Management System with Automated Billing Using Smart Trolley

Rai et al. (2019) explored the idea of developing a cloud-based inventory management system supplemented by automated billing. The authors suggested addressing long billing queues in retail stores using a smart trolley system.

### Literature Gap

According to the literature reviewed, it was found out that there were quite a number of software applications that concentrated either on managing inventories, billing or invoices alone. There were some software programs which were too costly, while other lacked security, customer management, or scalability. Very few software packages provided inventory management, billing, invoice creation, customer management, reporting capabilities, and security login.

### Relevance of Literature to Present Project

To counteract all these shortcomings, the present project known as Inventory and Billing Management System was created using MERN stack technology. This includes product management, inventory management, billing, invoice creation, customer management, reports and secured login among others.

From this literature review, it is clear that computerized business systems are very important in improving efficiency, eliminating manual errors and providing better customer services. From this limitation of past systems, the present project came up with an innovative approach.

# CHAPTER - 2

# IMPLEMENTATIONS

## 2.1 METHODOLOGY

For instance, in the case of development of the Inventory and Billing Management System, a modular and incremental method was used. This method entails splitting the whole software into several functional units such as authentication, product management, inventory management, billing, customers' management, and reports. All these modules are developed independently before being combined in order to produce the whole software.

The first step taken during the development of the system entailed analysis of the needs of the project. This included analyzing the various common challenges encountered while using the traditional inventory and billing management system. Some of the challenges resulting from the traditional manual approach include delays in bill preparation, inaccuracies in entries, duplicates, excessive paperwork, customer management, and reporting.

After the requirements gathering process is completed, the next stage is the planning process. This includes design of the architectural structure, navigational routes, roles of the users, interactions between modules, database, and layouts of several pages such as login, dashboard, product management, billing, and reports.

Following the completion of the product development stage was the transition to the product deployment stage. To build an official website with an interactive interface or frontend for the site, the React Library was utilized to complete this task. The backend of the site will consist of nodejs and express, with both working together to provide a mechanism for utilizing an application programming interface (API), authentication and business logic for the application. The database will be built using MongoDB to store user profiles and product information (data) (user profiles, product information, customer information and invoice information).

Once all modules were built and ready to be tested independently, all individual module functionality was completed before performing full application integration testing (integration of all modules).

Any defects or issues found during the testing process will be repaired and fixed. Functional testing, billing testing, login testing and database testing.

The next step in this product lifecycle will be to deploy the software into a cloud-based environment. The frontend was deployed using Vercel, the backend was deployed using Render, while MongoDB Atlas was used for hosting the database. Anyone can access the system as long as they have internet connectivity.

There were several advantages that came with the method adopted in the development of the system. This included simplifying the development process, improved code maintenance, reduced debugging efforts, scalability of the system, and future improvements like barcode scanning, e-commerce transactions, mobile applications, and analytics among others.

## 2.2 FRONTEND WORKFLOW

React technology was utilized in developing the frontend part of the Inventory and Billing Management System. The frontend represents the layer through which users interact with the application. It provides a platform for effective communication with the application. With the help of this section, the user can perform operations such as login, products, billing, customers, and reports.

On loading the application, the initial page opened is the Login page, where users enter usernames and passwords. After successful logging in, depending on the user's role, he/she is directed to the specific dashboard. Admin users will be redirected to the Admin Dashboard, while others to the Staff Dashboard.

Several pages are found in the frontend; some of these include the Product, Billing, Customer, Inventory, Dashboards, and Reports page. Navigation between pages is done via React router without having to reload the entire page.

Axios is what the frontend uses to make requests to the backend server. With every action taken by the user such as adding products, bills, viewing reports and managing customers, axios sends API requests to receive the information from the backend server. The obtained information is then rendered for display. In React, the stateful tasks are handled using hooks such as usestate and useEffect. These become very useful in storing dynamic data and automatic loading of information.

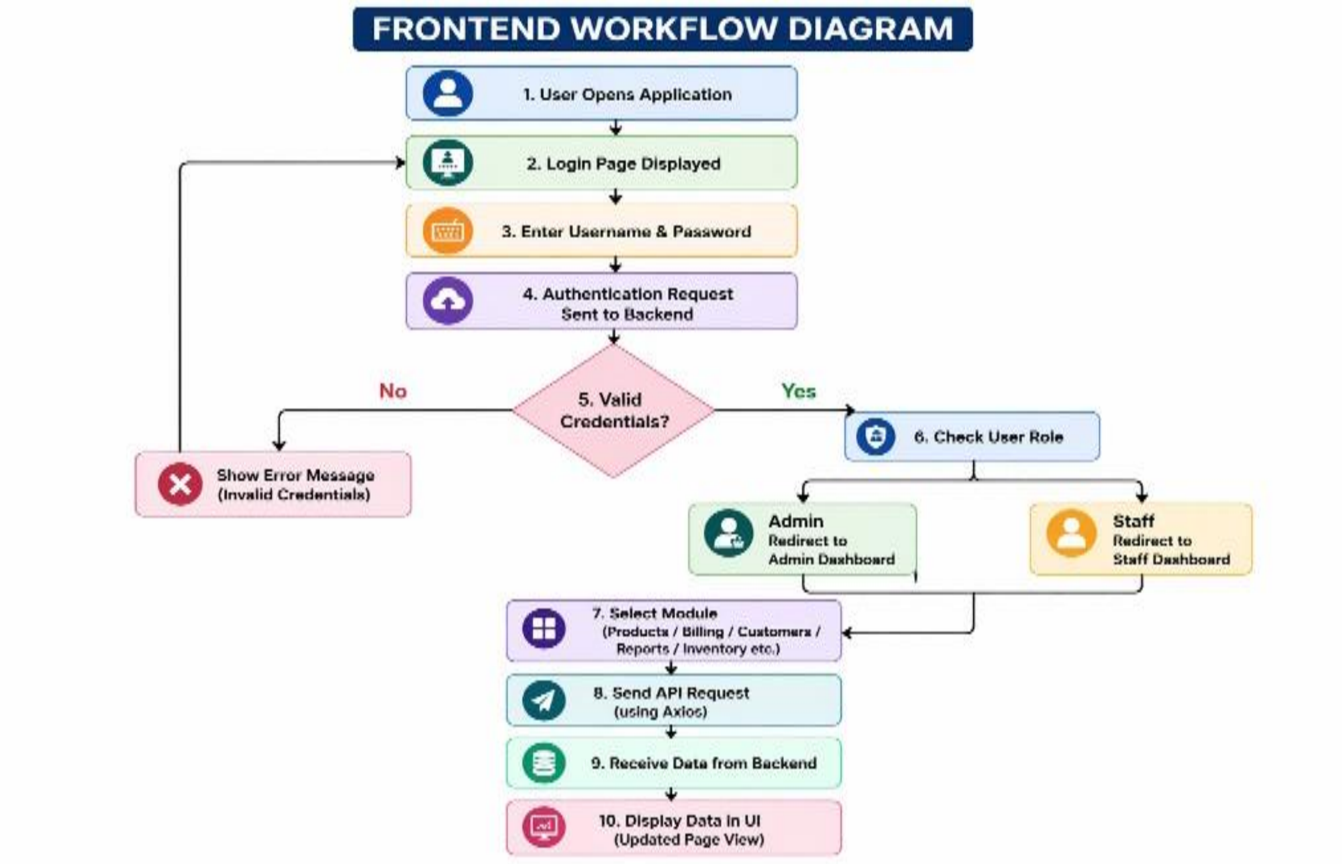



Fig1.Frontend Workflow

## 2.3 BACKEND WORKFLOW

Backend of the Inventory and Billing Management System is created using Node.js and Express.js frameworks. If the process has been completed and there are no errors in processing, then the necessary response data is sent back to the respective frontend application that initiated the request. The final format of this response data is in JSON format.

Every time someone interacts with the application from the client (frontend), whether they successfully log in, add a new product, generate an invoice or view a report, an API request gets sent out to the backend. This communicates the request to the backend through REST API calls created using Express.js routes depending on the configuration of the application. Express.js routes have been created for aspects such as

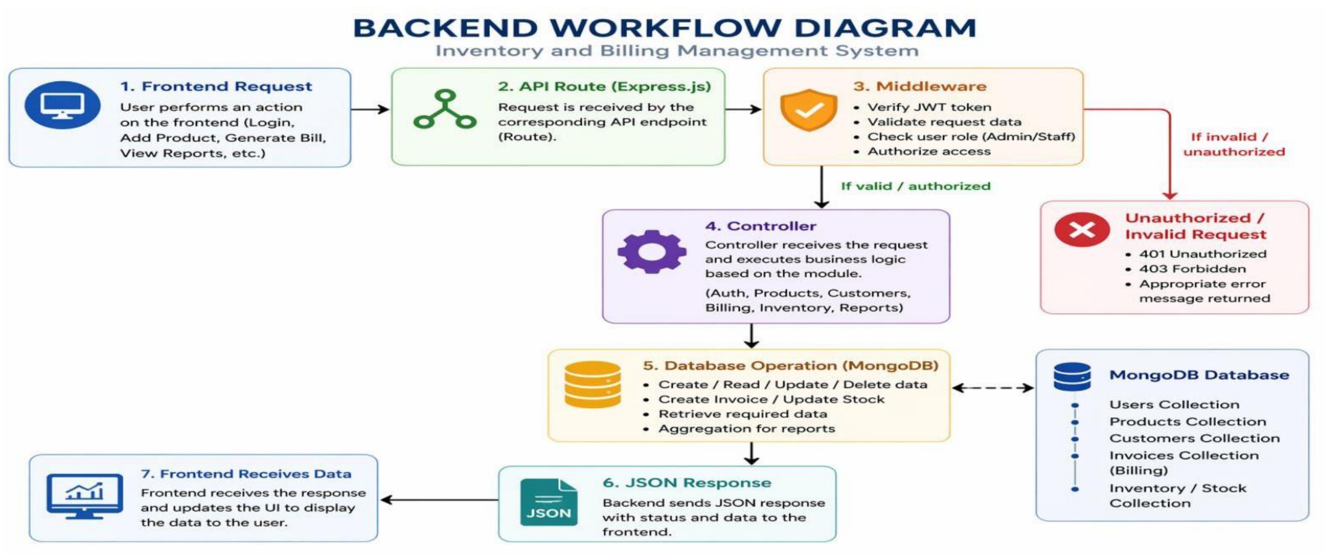
authentication, product creation, customer verification, invoicing, inventory control and reporting.
After the API call hits the Express.js route, middleware will process the request by verifying the JWT token through three separate middleware functions: (1) Verifying the JWT token that was passed in, (2) Validating the request data, i.e., ensuring that the data type being passed is correct, and (3) Verifying that the user's role is either 'admin' or 'staff'.

Once the request is verified, the request will be passed on to the appropriate controller action code that contains business logic associated with actions such as creating a product, updating inventory, getting customer information, generating invoices and performing calculations to generate reports.

The backend uses objects created by Mongoose to connect with MongoDB. Thus, depending upon the request, MongoDB will either be used to save, update, remove or retrieve data from the database.

After all of these processes are completed, if there are no issues with the processing of the request, the backend will send the appropriate JSON response containing the requested data back to the client/frontend.

Fig2.Backend Workflow



## 2.4 DATABASE WORKFLOW

The Inventory and Billing Management System works with a variety of operations – any combination of storing, retrieving, and managing data, on a daily basis. The source of data for this project is a MongoDB database that has been established as the database for this project and contains a document for each record (i.e., product, customer) in its corresponding collection. The connection to the database is made using Mongoose's model and schema function on the backend.

The front end can request information from the inventory and billing management system (via a user's session) and the back end will process the request and perform the relevant database action (insert/update)$/delete) and return the results of the database action to the front end to be displayed to the user.

The first time a user registers or enters the Inventory and Billing Management System (IBMS), their profile (user name, password, etc.) is saved to the Users collection of the database. Passwords are encrypted and the user role (admin or staff) is saved as part of their record.

When an administrator adds a new product to the Products collection of the database, the administrator inputs the product's name, category, cost, quantity, and a product description. Any edits made later to the product's information are saved and will be reflected in the database. If an administrator wants to delete a product from IBMS, they will remove the product from the Products collection in the database.

Details about customers, such as customer names, mobile numbers, addresses, and purchasing history, are saved in the Customers collection. The Customers collection is utilized during billing and customer management processes.

In the process of billing, details like products, amounts purchased, subtotals, taxes, discounts, total cost, payment methods, and dates are all saved in the Invoices collection. Once a sale has been made successfully, the stock levels for sold products will be reduced in the Products collection.
The creation of reports and dashboards will be done using data extracted from the invoices and products collections.

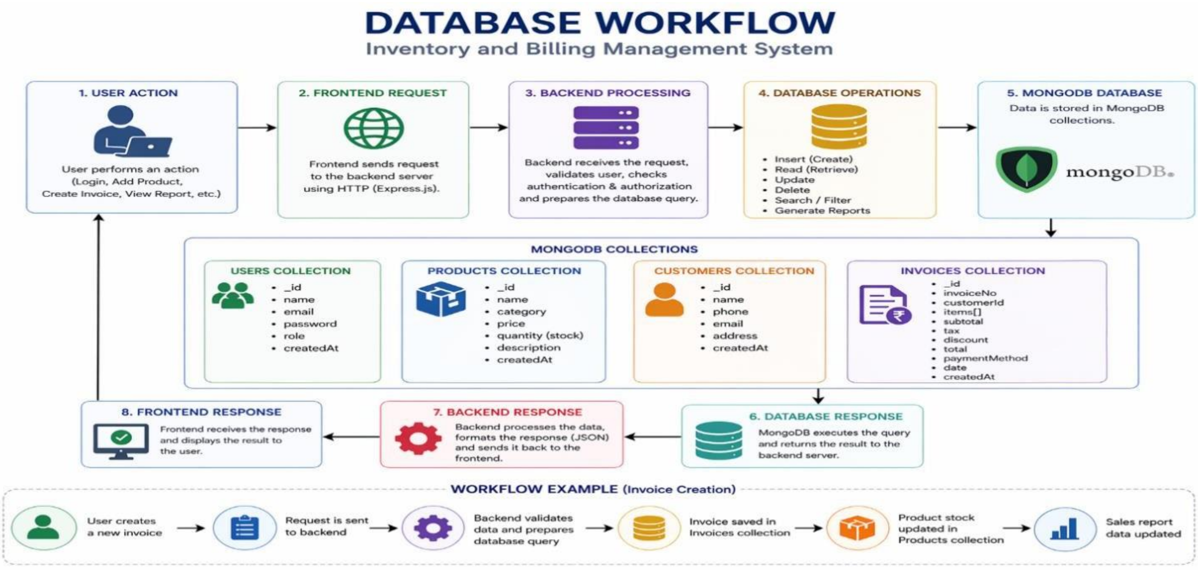



Fig3.Database Workflow

## 2.5 METHODS USED

Different contemporary approaches and techniques have been applied during the development of the Inventory and Billing Management System. They are used to optimize the performance of operations regarding managing data, performing secure authentication, conducting billing calculations, and interacting with the system.

CRUD technique was applied for managing products, customers, and users. CRUD represents four operations of creating data, reading existing data, updating information, and deleting unnecessary records.

The REST API approach was employed to facilitate interaction between the frontend and backend components. Different HTTP methods such as GET, POST, PUT, and DELETE were utilized to transfer information from React frontend to Node.js backend server.

JWT authentication approach was applied to implement secure logins into the application. Once users are successfully logged in, they are given tokens that allow accessing private routes of the app.

bcrypt approach was applied to encrypt passwords. When users set up their passwords, they are transformed into an encrypted form that is stored in the database.

Automatic bill amount calculation approaches were applied for the computation of subtotal, taxes, discounts, and total amounts. This eliminates human errors and ensures faster processing of bills.

Automatic stock updating approaches were applied to decrease the stock of products whenever there is a successful purchase. Through the implementation of search and filtering methods, the system allows for the quick finding of both customer and product records, thereby allowing for easy maintenance and less time required in creating a bill.

The system was designed using a Role-Based Access Control method, whereby administrative users have complete access and staff members have limited access only.

With the combination of all of these methods, an effective and reliable Inventory and Billing Management System has been created.

## 2.6 MODULE IMPLEMENTATION

These modules were used to achieve the purpose of performing certain functions by each module, which made the entire system more manageable, efficient, and systematic. Moreover, this helped in easy debugging, future updates, and feature integration.

**Login & Authentication Module**

The Login & Authentication module enables users to access the application. The user logs into the system with his email or username and password combination. After that, the user is authenticated through JWT authentication and bcrypt encryption. After this process, depending upon the user type, the user gains access to the admin or staff dashboard.

### Admin Dashboard Module

The Admin Dashboard module gives the entire control of the system. It displays all the statistics regarding total product quantity, total customers, sales report, stock availability, and many other things. Admin can control the products, customers, billing entries, and reports.

### Staff Dashboard Module

The Staff Dashboard module includes all those features related to daily tasks. The staff dashboard contains the features of billing, customer management, and inventory details.

### Product Management Module

Product Management module is used to update product information in the database. Users can add new products, edit product information, delete unwanted products, and search for products. Information related to product name, type, price, and quantity is stored in the database.

### Inventory Management Module

Inventory Management module is used to record the amount of stock available in the store. Each time a transaction occurs, the stock will be automatically updated. This feature ensures that there is no shortage of stock.

### Customer Management Module

Customer Management module is used to save customer data like customer name, contact number, address, and transaction history.
Using this information enables you to generate multiple invoices, as well as provide customer service.

**Billing and Invoice module**

The billing and invoice module of the application plays a key role in the system. Staff users can easily add items, enter quantities, apply discounts/taxes, and create invoices. All invoice amounts will automatically calculate and save to the database.

**Report module**

The report module provides the means to produce important analytics to help you gain insight into your business. You can create daily, weekly, monthly, and yearly sales reports, stock reports, and transaction reports. You can use these reports to analyze performance and make business decisions based upon the results.

## 2.7 SYSTEM ARCHITECTURE

The Inventory & Billing Management System (IBMS) is created on a three-tier architecture (also referred to as layers), specifically with regard to the MERN (MongoDB, Express, React, Node) technology stack. As represented in this layer cake structure, there are three layers to the IBMS: Presentation Layer (the UI); Application Layer (the Backend); and Data Layer (Database). Using this three-tiered layering approach, the project can be developed and scaled, be highly secure, and remain easy to maintain.

The Frontend of the IBMS is created using React, a technology for creating user interfaces. The Frontend is also what the Admins and Staff see when they use IBMS. Controls within the Frontend allow Admins and Staff to log into the system, view their dashboards, manage their products, bill their customers, and view their reports. In addition, the Frontend will allow Admins and Staff to enter data into the application by submitting forms that send information directly to the Backend via API calls.

The Backend of the application contains all of the business logic associated with the operation of the IBMS. The server-side application uses both Node.js and the Express Framework to provide server-side operations and database communication with MongoDB (the data layer of the application).

It receives requests from the frontend, verifies user credentials via JWT, performs billing services, manages stock products and customers, and produces reports.

Data Layer refers to the database where all the business data gets stored. MongoDB Atlas is a database solution for storing the collection like User, Product, Customer, Invoices. Data layer deals with storing, retrieving, updating, and generating reports on the data.

On any event that occurs by the user, the request from frontend will pass to backend using REST API. Backend will process the request, interact with the database, and send back the results to the frontend, which will be presented to the user.

It allows for secured communication, modular development, easy maintenance, and scalable development for future requirements of the application.

Layer of Architecture

Presentation Layer - React Frontend (Vercel)

Application Layer - Node.js + Express Backend

(Render) Data Layer - MongoDB Atlas Database

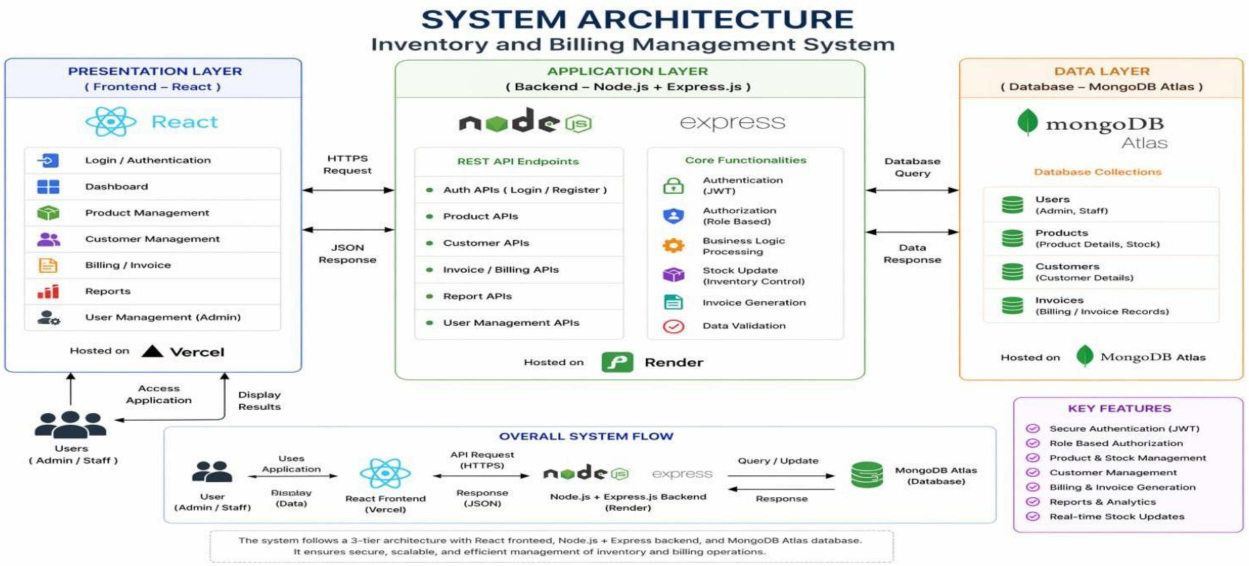



Fig4.System Architecture

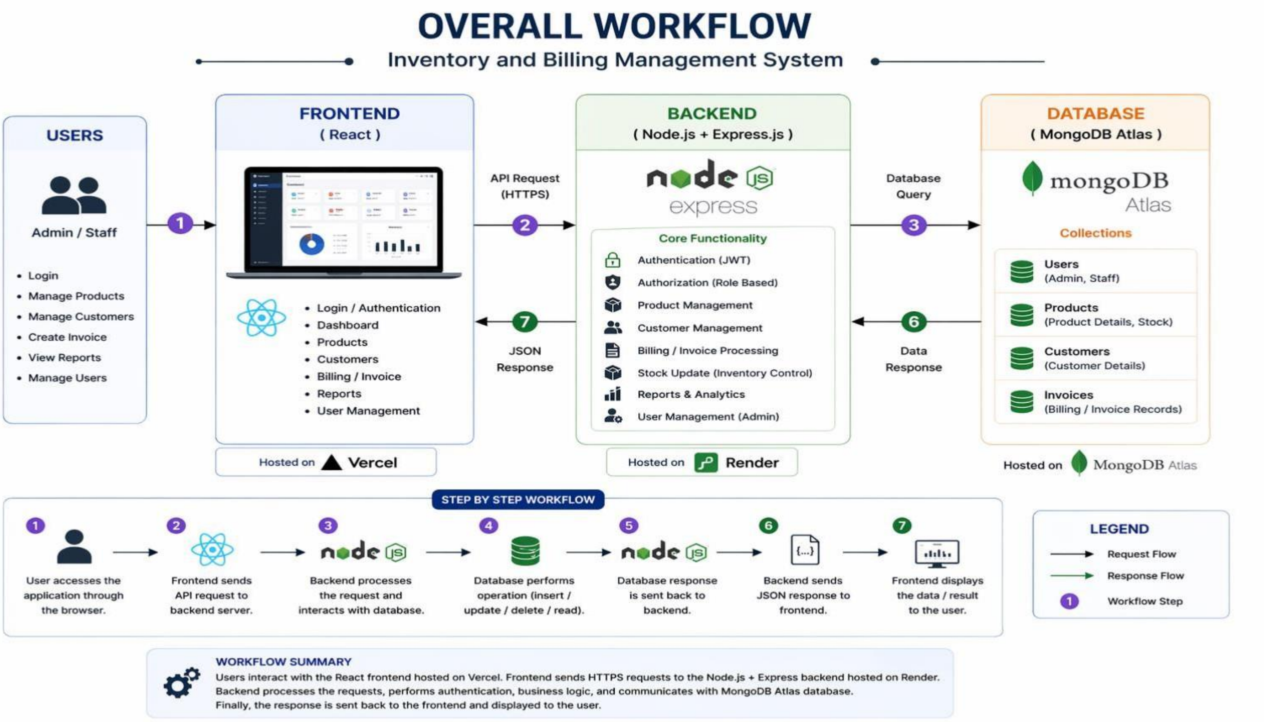



Fig5.Complete Workflow

## 2.8 TESTING AND VALIDATION

Various tests were performed to determine that the system operates efficiently, effectively, and securely. It ensures that all modules operate as expected, handle error situations, and meet business objectives.

Login test was performed to ensure that only users with valid credentials can log in while invalid credentials deny access. Other than login, various features including product management, customer management, billing, stock management, and reports were analyzed.

Test cases for product management included product addition, modification, deletion, and search operations. Billing test cases include product selection, quantity, discount, tax calculation, and generation of invoice. Before MongoDB storage worked properly, testing uses MongoDB to save, look-up, modify, and delete information in a database table. There was also some button test to check how the tab displays things in a user interface. After all problems had been fixed, the entire application was tested to check all design in order to confirm that the application provides  accurate output.

# CHAPTER-3

# TECHNOLOGIES USED

## 3.1 SYSTEM REQUIREMENTS

An inventory and billing systems-based toolset was developed using the MERN stack among various other tools, as they result in the development of applications for the web that are secure, highly scalable, and user-friendly. It included the following: MongoDB; Express.js; React.js; Node.js; JWT Authentication; Bcrypt; Axios.

**Hardware Requirements**

- Processing unit: Intel i3 minimum
- RAM: 4 GB or above
- Available space: 10 GB
- Internet access

**Software Requirements**

- OS: Windows, Linux, or Mac
- Front-end: React, HTML5, CSS3, and JavaScript
- Back-end: Node.js with Express.js
- Database: MongoDB or MongoDB Atlas
- Code Editor: Visual Studio Code
- API Testing Tool: Postman
- Web browser: Google Chrome or Microsoft Edge

## 3.2 TECHNOLOGY STACK

An inventory and billing systems-based toolset was developed using the MERN stack among various other tools, as they result in the development of applications for the web that are secure, highly scalable, and user-friendly. It included the following: MongoDB; Express.js; React.js; Node.js; JWT Authentication; Bcrypt; Axios.

**MongoDB**

MongoDB is an open-source NoSQL database, using documents that resembles JSON for data storage. We use MongoDB because it is highly flexible, fast, and horizontally scalable. In this project, MongoDB has been used to serve the following entities: users, products, customers, bills, and stock items. Some of MongoDB's key attributes are: flexible schema, fast querying, easy integration with Node.js, and cloud hosting.

**Express.js**

Express.js is a minimalist Node.js framework that comes equipped with useful functionality and tools for creating web and mobile applications. Express.js allows efficient routing and middleware implementation thus being perfect for the web applications and APIs. Advantages of Express.js include rapid development, scalability, clean URL routing and powerful backend performance.

**React**

React is a JavaScript library for developing interactive UI elements. The framework was chosen due to its component-based system, fast rendering, and code reusability. React is utilized on the frontend side to create all the pages, including the login page, dashboard, billing section, products, customers, and reports. Advantages of using React include responsive interface design, better UX/UI, code reusability, and easy-to-maintain code.

**Node.js**

Node.js is a cross-platform JavaScript runtime environment designed for the execution of server-side JavaScript code. This JavaScript framework was chosen due to its performance, asynchronous nature, and scalability. In this project, Node.js is utilized to run the backend server and handle API requests. Advantages of this framework include high performance, non-blocking approach, scalability, and cross-platform support.

### JWT Authentication

JSON Web Token is a stateless authentication method. It was chosen due to statelessness and the capability to securely store user sessions. JSON Web Tokens are utilized to create secure user sessions and ensure the protection of restricted routes during login. Also, the token allows role-based authentication to implement the difference between Admin and Staff roles.

### Axios

Axios is a library that helps in making API calls. Axios was used due to the ease it provided while making requests and handling errors. In this project, Axios was used to make login requests, fetch products from the database, update records, make bills, and create reports. Axios provides easy API integration, JSON parsing capabilities, request interception, and consistent error handling.

### Visual Studio Code

Small, fast, and very efficient, Visual Studio Code is an editor for your source code (for all parts – coding & file management), and is available for coding and managing files on your project. Visual Studio Code for this project was selected because of its small footprint, ability to debug, and numerous extensions available to extend your productivity. Visual Studio Code is used to build both the front end and back end of the application in this project. Some advantages of using Visual Studio Code include productivity, code formatting, debugging, and project management.

### Postman

Postman to allow testing through HTTP requests and responses. Postman in this project was used to perform the API tests against multiple endpoints on the back end: login, product, billing, and database.

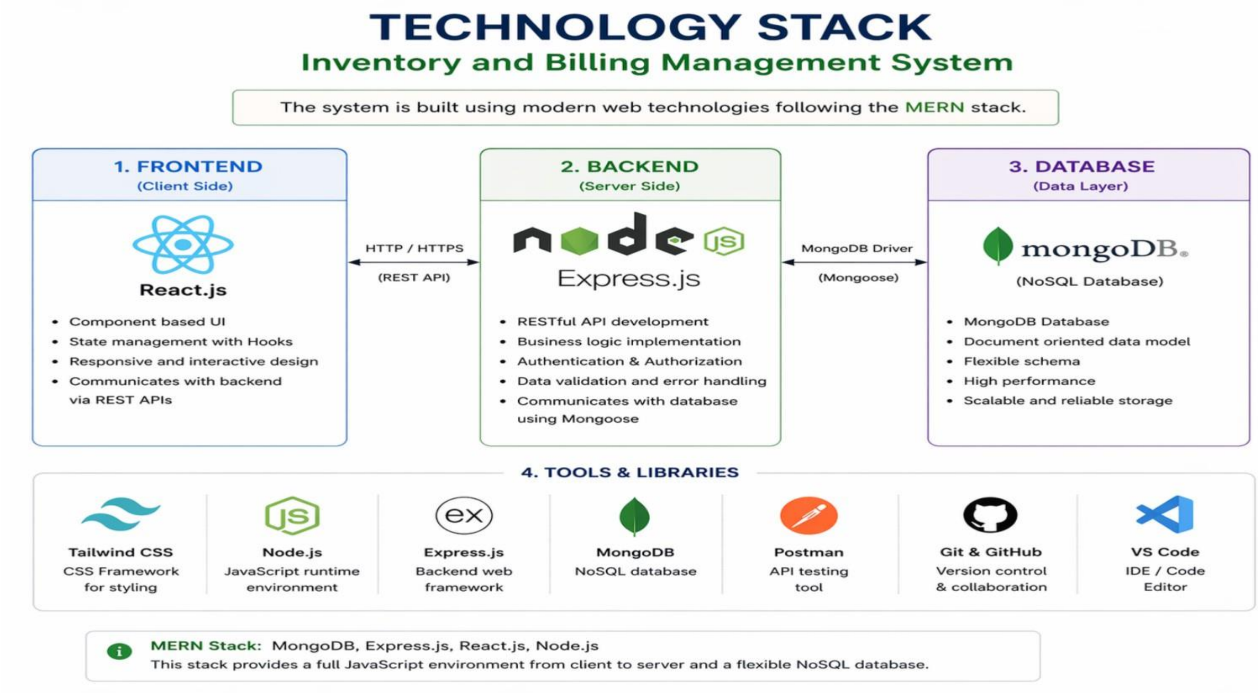



Fig6.Technology Stack

## 3.3 DATABASE DESIGN

The data contained within the Inventory & Billing Management System is organized and held by a well-designed database. The type of database that is used is MongoDB, which is a NoSQL database. Therefore, data is stored in JSON-like documents within collections - allowing for flexibility with schema and providing high speeds with easy scalability, making it suitable for web applications.

Some of the modules supported by the database include authentication, product and customer management, billing, inventory management, and reporting. Schemas, validations, and models for each collection are provided by Mongoose.

### User’s Collection

In this collection, you will get all the information related to user credentials, be it Admin or Staff.

The fields to focus on here are username, password, and role. Role clearly indicates whether the user is an Admin or Staff.

**Product Collection**

This collection holds all the information regarding products which are helpful to monitor the product inventory and pricing.

Main fields in this collection are product name, product price, and available units. With the generation of each bill, the number of products gets automatically deducted.

**Customer Collection**

In this collection, we will have all details of customers which are useful for generating bills. Name, contact number, email, and created at are some basic fields of customer collection.

We want to keep a good history of our customers to ease out the payment process in the future.

**Invoice Collection**

This collection holds all the information regarding billing. Subtotal, taxes, discount, total, and created at are some main fields of Invoice Collection. Invoices contain items arrays which list all the items purchased in that invoice. This list contains product id, product name, price, and quantity.

**Connection Between Collection**

The collection in the system is well-connected logically in a network. The user can work with products and billing tasks, where products are retrieved when billing products are included in invoice items. Customer details are stored in a separate collection for proper record management. When creating an invoice, the inventory levels for the purchased product will be reduced.

**Advantages of the Database**

- Proper data management
- Minimal data repetition
- Efficient data search
- Ease of maintenance
- Scalability
- Record security

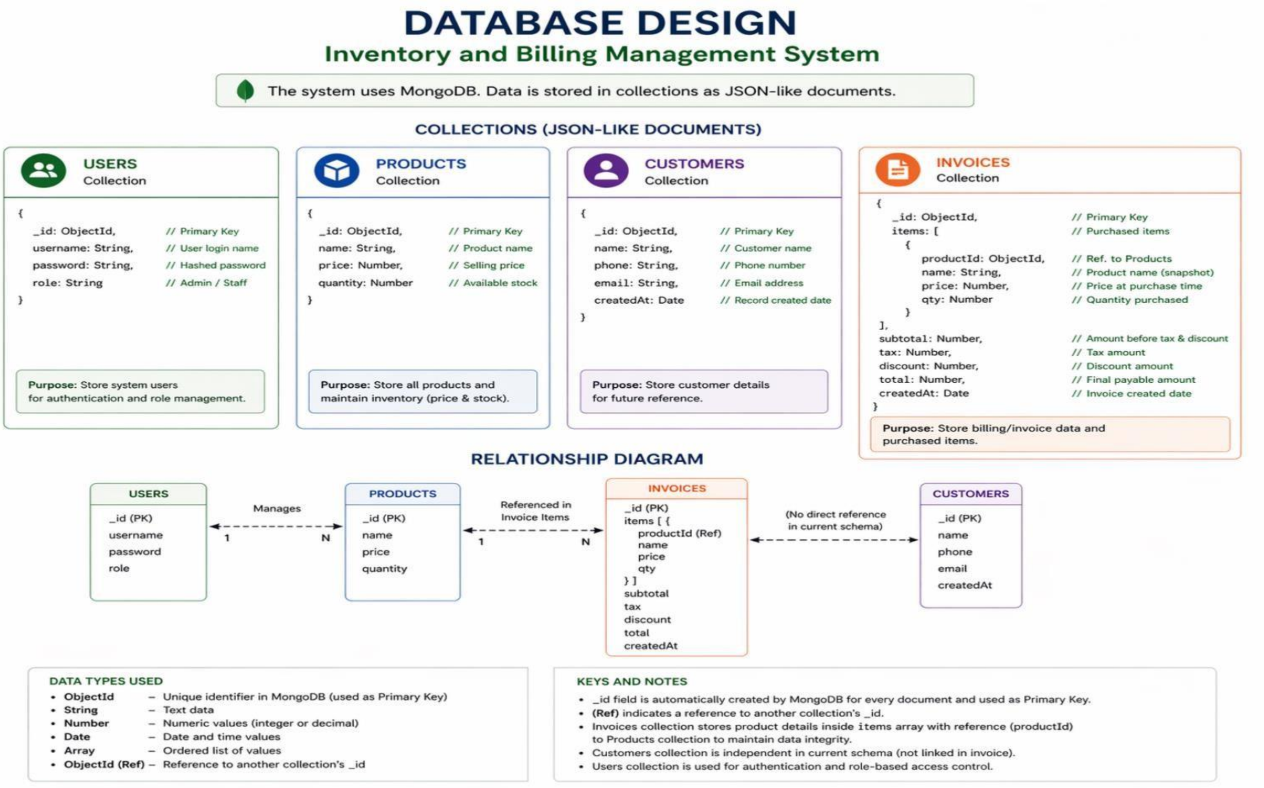


Fig7.Database Design

## 3.4 IMPORTANT FEATURES OF THE SYSTEM

Some of the vital features available in the Inventory and Billing Management System are discussed below, which contribute to efficient business activities. These features include those developed to enhance speed of billing process, inventory management, data accuracy, security, and management of users.

### User Authentication

It is possible to log in to the system securely with the right credentials. It is only those users who have been registered to use the application that can use the system.

### Role-Based Access Control

The project allows users to operate based on their roles, including Admins and Staff. While admins have complete control over products, users, and generation of reports, staff members are able to perform billing functions.

**Product Management**

With this module, the system offers the ability to add new products, update, delete, and view existing products. It is also easier to manage product price and inventory using this function.

**Billing and Invoicing**

With QuickBooks, you have access to a fast way to create an estimate and generate an invoice. You can easily add an item to your invoice or include the total cost of an invoice, apply sales tax and/or discounts and then print the bill directly. This not only helps improve the accuracy of your invoice but also saves you time.

**Customer Management System**

In addition, you can enter customer names, phone numbers, and email addresses into the QuickBooks database so that you can easily generate invoices for customers who make repeat purchases and maintain accurate records on the customer's history.

**Generating Reports**

QuickBooks also has the ability for you to keep information on all previous invoices and transaction history so that you can create reports and perform analysis on this information to assist in the decision-making process for your business.

**Responsive Interface**

The user interface at the front end is made in such a way that it responds efficiently to the operations performed.

**Password Protection**

Password protection makes sure that only authorized persons have access to critical business information.

## 3.5 DEPLOYMENT PLATFORMS

Deployment of the Inventory and Billing Management System occurred using cloud-based platforms which enabled accessibility of the application online, improved security, and simplified maintenance.

The deployment was performed in three stages, including frontend hosting, backend server hosting, and database hosting.

### Frontend Deployment – Vercel

The frontend portion of the project developed using React was hosted on Vercel. Vercel is a cloud service that can be used to deploy React projects as it features fast deployment and other advantages. Vercel allows users to easily access the application through their web browsers anywhere.

### Backend Deployment – Render

The backend server portion of the project which includes the development of APIs and other server-side operations using Node.js and Express.js frameworks was hosted using Render. This cloud platform offers security, support for environment variables, automatic deployment, and always available servers.

### Database Deployment – MongoDB Atlas

The database of the project was hosted using MongoDB Atlas. MongoDB Atlas is a cloud-based service that can be used to host MongoDB databases online. MongoDB Atlas provides remote access, automated backups, and monitoring among other useful features.

### Benefits of This Deployment Approach

- Online accessibility of the application
- Separation of hosting services
- Simplified maintenance and updates
- High security and reliability
- Better scalability and performance
- Decreased local dependency

# CHAPTER – 4

# ALIGNMENT WITH SUSTAINABLE DEVELOPMENT GOALS (SDGs)

## 4.1 INTRODUCTION TO SUSTAINABLE DEVELOPMENT GOALS (SDGs)

The Sustainable Development Goals (SDGs) were introduced by the United Nations in 2015 as a global framework to address challenges such as poverty, inequality, education, economic growth, and sustainable development. These 17 goals aim to ensure balanced progress across social, economic, and environmental dimensions by the year 2030.

For engineering and software development, SDGs provide a guideline to build systems that are not only efficient but also socially beneficial and sustainable. Digital platforms, automation systems, and data-driven applications contribute significantly to improving accessibility, productivity, and transparency.

The 17 SDGs include goals related to education, innovation, economic growth, sustainability and governance.

1.No poverty

2.Zero Hunger

3.Good Health and Well-bei.ng

4.Quality Education

5.Gender Equality

6.Cle.an Water and Sanitation

7.Affordable and Clean Energy

8.Decent Work and Economic Growth

9.Industry, Innovation, and Infrastructure

10.Reduced Inequalities

11.Sustainable Cities and Communities

12.Responsible Consumption and Production

13.Climate Action

14.Life Below Water

15.Life on Land

16.Peace, Justice and Strong Institutions

17.Partnerships for the Goals

## 4.2 MAPPING OF RELAVENT SDGs AND BREIF DESCRIPTION

**SDG 9: Industry, Innovation, and Infrastructure**

- Inventory and Billing Systems represent digital innovation in modern business practices.
- They help industries transition from manual to automated processes.
- These systems strengthen technological infrastructure within organizations.
- They encourage the adoption of software solutions for better operational control.
- Such innovations enhance overall industrial efficiency and competitiveness.

**SDG 12: Responsible Consumption and Production**

- These systems help track stock levels accurately, preventing overstocking and shortages.
- They reduce wastage by monitoring product movement and expiry dates.
- Businesses can plan production and purchasing more responsibly using real-time data.
- They promote efficient use of resources through proper inventory management.
- This leads to sustainable consumption patterns and reduced environmental impact.

**SDG 8: Decent Work and Economic Growth**

- Inventory and Billing Management Systems improve business efficiency by automating daily operations.
- They reduce manual errors in financial transactions and inventory tracking.
- These systems help businesses maintain accurate financial records, supporting better decision-making.
- They contribute to business growth by saving time and operational costs.
- By improving productivity, they support job stability and economic development.

# CHAPTER - 5

# CONCLUSION

## 5.1CONCLUSION

The design and implementation of the Inventory and Billing Management System is intended for creating an easy-to-use solution that will help manage the inventory, billings, and clients' accounts. This software is developed using the MERN technology stack that includes React.js as a framework for building front-end components, and Node.js and Express.js used for backend. The database is represented by MongoDB.

The main functionality of the software involves secure login, role-based system access (Admin or Staff), products and clients management, invoicing and automatic inventory update that minimize manual work, increase speed of billing operations, minimize errors, and provide accurate accounting.

In general, with its simple UI, the software can be recommended for use in a small business. In conclusion, all the project goals were successfully reached.

## 5.2 FUTURE SCOPE

Further improvement of the project could be done in future through various ways. Barcode scan feature can be used for quick searches as well as quick billing purposes. Digital payment gateway integration can be included in order to enable the transaction through online payment methods such as UPI, cards, wallets, and others.

Invoice notifications via email and SMS services can also be provided to maintain better contact with the customers. Various analytical tools such as graphs and other features can prove beneficial for decision-making by the business owner. The mobile app can be created for using the application on the go through smartphones and tablets.

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